Topic 20 — Biodiversity Conservation Methods

Covers syllabus (click to expand)

In-situ (स्थान पर) Conservation (स्व-स्थाने संरक्षण) | Ex-situ (स्थानांतरित) Conservation (बाह्य-स्थाने संरक्षण) | Gene Bank | Seed Bank (बीज बैंक) | Botanical Garden (वनस्पति उद्यान) | Zoological Park

Sources baked in: NCERT Bio 12 Ch 15; CBD 1992; WPA 1972; BDA 2002; ICAR-NBPGR; NBAGR; CZA; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — in-situ vs ex-situ traps; NBPGR/NBAGR location; facility ↔ method matching **Last verified:** August 2026

Current Affairs (this topic)

- 30×30 / Kunming–Montreal CA sits here with in-situ vs ex-situ method traps.
- Sacred grove and seed-bank examples stay teaching baselines.

20.1 In-situ Conservation

Identity: On-site | Natural habitat | Preferred primary strategy | WPA 1972 protected areas

- **In-situ conservation** protects species **where they naturally live**, keeping habitat, food webs, pollinators, and competitors intact.
- NCERT and global policy treat in-situ as the **preferred primary strategy** because it preserves **ecosystem integrity**, **natural selection**, and **genetic diversity** (आनुवंशिक विविधता) in real conditions.
- **National Parks** offer strict habitat protection under **Wildlife (Protection) Act, 1972**, with human activity restricted. **Dudhwa NP (UP)** is a Terai (तराई) in-situ example.
- **Wildlife Sanctuaries** protect habitat with slightly more flexible human use than NPs but remain **in-situ**. **Katerniaghat WLS (UP)** protects gharial and Gangetic dolphin in natural rivers.
- **Biosphere Reserves** follow the UNESCO **MAB** model with **core**, **buffer**, and **transition** zones, combining strict protection with sustainable community use.
- **Conservation Reserves** and **Community Reserves** (WPA amendments) protect habitat patches with community participation.
- **Sacred groves** (deorai, sarna (सरना), kavu) are traditional **in-situ** community forests protected by local belief systems.
- **Ramsar wetlands** and **marine protected areas** are specialised **in-situ** tools for aquatic and coastal ecosystems.
- **Wildlife corridors** link fragmented habitats at landscape scale. The **Terai Arc Landscape** connects Dudhwa–Corbett habitats.
- **Limitation:** In-situ alone cannot save species whose habitat is completely destroyed or whose wild population (जनसंख्या) is critically small without ex-situ backup.

PYQ — UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None of the above

Show answer

Ans: A — NPs protect wild populations **in natural habitat** = in-situ.

Method	Type	India example
National Park	In-situ	Dudhwa (UP)
Wildlife Sanctuary	In-situ	Katerniaghat (UP)
Biosphere Reserve	In-situ	Similipal
Sacred Grove	In-situ	Tribal (आदिवासी) UP belts
Seed Bank	**Ex-situ**	NBPGR Delhi

> **Logic:** UPPCS 2019 pattern — "Which is in-situ?" → only **NP/sanctuary/BR/sacred grove**. Zoo, seed bank, botanic garden = **ex-situ distractors**.

20.2 Ex-situ Conservation

Identity: Off-site | Supplementary backup | Insurance against extinction | Cannot replace in-situ

- **Ex-situ conservation** removes or copies genetic material **outside the natural ecosystem** — zoos, seed banks, gene banks, captive breeding, tissue culture.
- It is **supplementary**, not a substitute. Ex-situ cannot replicate full ecosystem complexity, predator–prey dynamics, or natural selection.
- **When essential:** critically small wild populations, **habitat already lost**, **reintroduction stock** needed (gharial, vulture), or **Extinct in the Wild (EW)** species surviving only in collections.
- **Major ex-situ types:** zoological parks, botanical gardens, seed banks, gene banks, captive breeding centres, tissue culture, cryopreservation.
- **Genetic bottleneck risk:** small captive groups lose diversity through inbreeding — zoos rotate breeding pairs and maintain studbooks.
- **Reintroduction link:** ex-situ breeding succeeds only if **in-situ habitat** is restored. Cheetah reintroduction required **Kuno NP** habitat preparation alongside captive stock.

- **IUCN (आईयूसीएन) Red List "Extinct in the Wild (EW)"** marks species surviving only in zoos, botanic gardens, or seed banks with no viable wild population.
- **CBD + BDA 2002:** India regulates access to genetic resources; ex-situ collections at NBPGR/NBAGR support national biodiversity obligations.

PYQ — UPPCS Prelims 2018, Q54

Which of the following is NOT an example of insitu conservation strategy?

A. Biosphere reserve

B. Botenical garden

C. National Park

D. Sacred groves

Show answer

Ans: B — Botanical garden = **ex-situ** living collection. All others = in-situ.

Feature	In-situ	Ex-situ
Location	Natural habitat	Captive/stored
Priority	**Primary**	**Supplementary**
Examples	Dudhwa NP, sacred grove	Delhi Zoo, NBPGR
trap	NP, sanctuary, BR	Zoo, seed bank, botanic garden

> **Logic:** "Zoos are the best/primary form of biodiversity conservation" = **FALSE**. In-situ protected areas remain primary.

20.3 Gene Bank

Identity: Ex-situ genetic repository | NBPGR Delhi (plants) | NBAGR Karnal (animals) | Cryo at -196°C

- A **gene bank** stores **multiple forms of genetic material**: seeds, pollen, tissue cultures, DNA, animal semen, and embryos.
- **ICAR-NBPGR** at **Pusa Campus, New Delhi** (est. **1978**) is India's national **plant** gene bank. It holds one of the world's largest plant collections.
- The **National Gene Bank** at NBPGR (est. **1996** unit) is the long-term **base collection** stored at **-18°C to -20°C** for orthodox seeds, with cryo units for recalcitrant material.
- **NBAGR, Karnal (Haryana)** (est. **1984**) conserves **animal** genetic resources — semen, embryos, and DNA of indigenous livestock breeds (Sahiwal, Gir (गीर), Red Sindhi).
- **Cryopreservation** uses **liquid nitrogen at -196°C**, stopping biological activity for decades-long storage of embryos, sperm, pollen, and tissue.

- **Active collection** holds frequently accessed seeds for breeders. **Base collection** is long-term duplicate storage with minimal disturbance.
- Stored seeds lose viability over time and must be **periodically regenerated** by germination and re-collection.
- **Trap: NBPGR vs BSI** — NBPGR **stores** genetic resources (Delhi). **BSI** does plant **survey/taxonomy** (Kolkata).
- **Trap: NBPGR vs NBAGR** — **NBPGR = plants (Delhi)**; **NBAGR = animals (Karnal)**.

Institution	Location	Conserves
ICAR-NBPGR	New Delhi	Plant seeds, pollen, tissue, DNA
National Gene Bank	New Delhi (NBPGR)	Long-term base collection
NBAGR	Karnal, Haryana	Livestock semen, embryos, DNA
Svalbard Global Seed Vault	Norway	Duplicate global seed backup

> **Logic:** "National gene bank for animal genetic resources is NBPGR" = **FALSE** — animals = **NBAGR Karnal**.

20.4 Seed Bank

Identity: Ex-situ | Seeds only | Orthodox = cold storage | Recalcitrant = mango/coconut trap

- A **seed bank** stores **seeds only** (narrower than a gene bank). Seeds are dried to **3–7% moisture**, sealed airtight, and kept at **-18°C to -20°C**.
- **Orthodox seeds** (wheat, rice, maize, millets, pulses) tolerate desiccation and cold — suitable for **decades of storage**.
- **Recalcitrant seeds** (mango, jackfruit, coconut, sal) **cannot tolerate drying** and need **botanical garden living collections** or **cryopreservation** instead of conventional seed banking.
- **NBPGR** operates the **National Active Collection** (breeder access) and **National Gene Bank** (long-term base store) with **4 lakh+ accessions** including landraces and crop wild relatives.
- **Svalbard Global Seed Vault (Norway, 2008)** stores **duplicate copies** of national collections as Arctic "doomsday backup" — not ownership transfer.
- **Regeneration** is mandatory when germination rates fall — gene banking is dynamic, not store-and-forget.
- **Community seed banks** (on-farm) preserve landraces in active cultivation — different from **formal ex-situ cold storage** at NBPGR. Notes usually mean NBPGR-style cold storage.

Type	Storage	Examples
Orthodox	Conventional seed bank (-18°C)	Wheat, rice, chickpea
Recalcitrant	Living collection or cryo	Mango, coconut, jackfruit

Type	Storage	Examples
Feature	Seed Bank	Gene Bank
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Material	Seeds primarily	Seeds + DNA + tissue + sperm + embryos
Scope	Narrower	Broader
Both are	**Ex-situ**	**Ex-situ**

> **Logic:** "Seed bank is in-situ conservation" = **FALSE**. UPPCS 2019 NOT-in-situ MCQs use seed bank as the ex-situ answer.

20.5 Botanical Garden

Identity: Ex-situ living plants | Indian Botanic Garden Howrah 1787 | BSI administers | Not same as BSI mandate

- **Botanical gardens** maintain **living plant collections** outside wild habitat for conservation, research, education, and display.
- Unlike seed banks (dormant storage), gardens keep **living plants** — essential for **recalcitrant-seed species** and **Extinct in the Wild** plants.
- The **Acharya Jagadish Chandra Bose Indian Botanic Garden** at **Howrah, West Bengal** (est. **1787** by Colonel Robert Kyd) is the **largest and oldest** botanic garden in India, famous for the **Great Banyan Tree**.
- It is administered by **Botanical Survey of India (BSI)** (भारतीय वनस्पति सर्वेक्षण). BSI is a **survey organisation** — it is not synonymous with "botanical garden" as a conservation method.
- **Other major gardens:** Lalbagh (Bengaluru, 1760), **NBRI garden (Lucknow)**, Empress Garden (Pune (पुणे)).
- **BGCI** links **3500+ gardens** globally for coordinated ex-situ plant conservation.
- **Limitation:** living collections hold limited genetic diversity, artificial conditions differ from wild ecology (पारिस्थितिकी), and maintenance is costly.

Garden	Location	Notable
Indian Botanic Garden	Howrah	1787; Great Banyan; largest in India
Lalbagh	Bengaluru	1760; heritage trees
NBRI Garden	Lucknow	Medicinal plant research

> **Logic:** UPPCS in-situ MCQs always list **botanical garden** as an **ex-situ** distractor alongside zoo and seed bank.

20.6 Zoological Park

Identity: Ex-situ captive animals | CZA under MoEFCC | Delhi Zoo 1959 | Captive breeding for recovery

- **Zoological parks (zoos)** hold animals in **enclosed, managed environments** — classic **ex-situ** conservation removed from free-ranging wild habitat.
- **Roles:** conservation breeding, public education, scientific research, and genetic reservoir for reintroduction when habitat is secured.
- **Central Zoo Authority (CZA)** (केंद्रीय चिड़ियाघर प्राधिकरण) was created under the **Wildlife (Protection) Amendment Act, 1991** and functions under **MoEFCC**. It grants zoo recognition (मान्यता), sets standards, and coordinates breeding programmes.
- The **National Zoological Park (Delhi Zoo)** in **New Delhi** (est. **1959**) is a flagship ex-situ facility for endangered Indian species.
- **Species recovery centres** (also ex-situ): vulture breeding centres (Pinjore, Rani, Buxa), **Kukrail Crocodile Centre (Lucknow, UP)**, gharial breeding for river release.
- **Trap: zoo vs sanctuary** — zoo = **captive ex-situ**; sanctuary = **wild animals in natural habitat (in-situ)**.
- **Limitations:** inbreeding in small groups, loss of wild survival skills, stereotypic behaviour — zoos **cannot replace** habitat-level in-situ protection.

Facility	Location	Type
National Zoological Park	New Delhi	Zoo (ex-situ)
CZA	New Delhi	Regulator (MoEFCC)
Kukrail Crocodile Centre	Lucknow, UP	Captive breeding (ex-situ)
Katerniaghat WLS	UP	Sanctuary (**in-situ** contrast)

> **Logic:** Every UPPCS in-situ question lists **zoological park** as ex-situ. Match: Zoo = ex-situ | NP/Sanctuary = in-situ.